



Tools and Technologies for Internet Programming Final Project Report On

**Sumtasinia Medica : A Web-Based Disease Awareness and Health
Information Platform**

Course Code : CSE-3532

Course Title : Tools and Technologies for Internet Programming

Submitted By

Sumaya Rahman (C233479)

Mumtasin Alam Suporna (C233487)

Submitted To:

Ahasanul Kalam Akib

Adjunct Lecturer, Department of Computer Science and Engineering
International Islamic University Chittagong

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Problem Statement :

In many parts of Bangladesh, especially in remote, hilly, and disaster-affected regions, people face difficulties accessing reliable healthcare services due to the shortage of expert doctors. Additionally, individuals living in rural areas often lack sufficient knowledge about various diseases, their symptoms, and available treatments. As a result, they struggle to get accurate health information and timely guidance. There is a need for a centralized, easily accessible platform that provides trustworthy and comprehensive health-related information in one place. So we will build a website which focuses on disease awareness. Our website will focus not only on creating awareness but also will provide necessary information and help to disadvantaged people.

Objectives:

- Creating an intuitive website that connects patients and doctors, with a clean and professional design for easy navigation. Focusing on simplicity and trustworthiness to appeal to users seeking reliable health information.
- Developing an **Articles Section** with detailed, credible information on common diseases (e.g. symptoms, causes, prevention, treatments), sourced from trusted references like WHO or CDC.
- Including a **Search and Filter Option** to let users find articles by disease name, category, or keyword quickly.
- Implementing separate sign-up and login systems for **patients** and **doctors**, ensuring secure access with encrypted credentials.
- Building an **Appointment Booking System** where patients can view doctors' profiles (name, specialty, availability) and book appointments via a simple calendar or form.
- Creating a **Dashboard** for Doctors to view upcoming appointments, access patient medical histories (shared by patients), and manage their availability and for patient to upload, view, and manage medical history.
- Adding a **Chatbot** to help users find disease articles or navigate the site.
- Developing an **Admin Panel** to manage users (patients/doctors), articles, and appointments.

Key Features :

- User Authentication : Secure login for patients and doctors
- Appointment Booking System
- Upload, view, and manage patient medical history
- Dashboard for doctors
- Dashboard for patients
- Articles related to diseases
- Search and filter option
- Upload hospital dataset for research
- Chatbot
- Admin Panel

Technology Stack :

Frontend : ReactJS

Backend : Node.js, ExpressJS

Database : SQL/MongoDB

Authentication: Firebase

Additional Tools: REST API, GEMINI API

Target Users :

- Users who value online access to healthcare services, including remote consultations, for convenience and time-saving.
- Doctors and healthcare professionals managing appointments.
- Users who want to educate themselves about different diseases with credible information

Expected Outcomes:

The expected outcome of this project is it will create an easy and reliable access to healthcare services for patients. In addition, it will create a bridge between doctors and patients. Researchers can use this website for their future research using datasets. Moreover, anyone can get to know about existing diseases and use this website as their encyclopedia. Lastly, it will ensure user experience through a reliable, digital healthcare solution.

Future Scope:

1. Medical Image Processing for Automated Diagnosis Assistance.
2. Multimodal Deep Learning Models for Disease Prediction.
3. Wearable Device Data Integration for Health Tracking.
4. Mobile App Development.
5. Multi-Language Support for Different Regions.